Program:

```
#include <stdio.h>
#define SIZE 5
int items[SIZE];
int front = -1, rear = -1;
int isFull()
{
 if ((front == rear + 1) || (front == 0 && rear == SIZE - 1))
 return 1;
 return 0;
}
int isEmpty()
{
 if (front == -1)
 return 1;
 return 0;
}
void enQueue(int element)
{
 if (isFull())
 printf("\nQueue is full!!\n");
 else
 {
 if (front == -1) {
 front = 0;
 rear = 0;
 } else {
 rear = (rear + 1) % SIZE;
 }
 items[rear] = element;
 printf("\nInserted -> %d", element);
 }
}
int deQueue()
{
 int element;
 if (isEmpty())
 {
 printf("\nQueue is empty!!\n");
 return -1;
 } else {
 element = items[front];
 if (front == rear)
```

```
 {
 front = -1;
 rear = -1;
 } else {
 front = (front + 1) % SIZE;
 }
 printf("\nDeleted element -> %d\n", element);
 return element;
 }
} void display()
{
 int i;
 if (isEmpty())
 printf("\nEmpty Queue\n");
 else
 {
 printf("\nFront -> %d", front);
 printf("\nItems -> ");
 for (i = front; i != rear; i = (i + 1) % SIZE)
 printf("%d ", items[i]);
 printf("%d ", items[i]);
 printf("\nRear -> %d\n", rear);
 }
}
int main()
{
 deQueue();
 enQueue(1);
 enQueue(2);
 enQueue(3);
 enQueue(4);
 enQueue(5);
 enQueue(6);
 display();
 deQueue();
 display();
 enQueue(7);
 display();
 enQueue(8);
 return 0;
}
```

Output:

Queue is empty!!
Inserted -> 1
Inserted -> 2
Inserted -> 3
Inserted -> 4
Inserted -> 5
Queue is full!!
Front -> 0
Items -> 1 2 3 4 5
Rear -> 4
Deleted element -> 1
Front -> 1
Items -> 2 3 4 5
Rear -> 4
Inserted -> 7
Front -> 1
Items -> 2 3 4 5 7
Rear -> 0
Queue is full!!